nature portfolio

https://doi.org/10.1038/s41586-025-08801-w

Supplementary information

Long-distance coherent quantum communications in deployed telecom networks

In the format provided by the authors and unedited

Supplementary Material for: Long-Distance Coherent Quantum Communications in Deployed Telecom Networks

Mirko Pittaluga[1*], Yuen San Lo[1], Adam Brzosko[1], Robert I. Woodward[1], Davide Scalcon[1], Matthew S. Winnel[1], Thomas Roger[1], James F. Dynes[1], Kim A. Owen[1], Sergio Juárez[1], Piotr Rydlichowski[2], Domenico Vicinanza[3,4], Guy Roberts[3] & Andrew J. Shields[1]

[1] *Toshiba Europe Limited, 208 Cambridge Science Park, Cambridge CB4 0GZ, UK*

[2] *Poznan Supercomputing and Networking Center, Jana Pawła II 10, 61-139 Poznan, PL*

[3] *GÉANT Vereniging, Hoekenrode 3, 1102BR Amsterdam, NL*

[4] *School of Computing and Information Science, Anglia Ruskin University, Cambridge, UK*

[*] *mirko.pittaluga@toshiba.eu*

S1. COMPARISON WITH STATE-OF-THE-ART PRACTICAL FIELD TRIAL EXPERIMENTS

The state-of-the-art field trial Quantum Key Distribution (QKD) experiments using non-cryogenically cooled detectors, reported for comparison in fig. 5 of the main text, are listed in chronological order and linked to their respective references as follows: [a] → [1], [b] → [2], [c] → [3], [d] → [4], [e] → [5], [f] → [6], [g] → [7], [h] → [8], [i] → [9], [l] → [10].

S2. CHARACTERISATION OF THE FIBRE LINKS

The experiment's location was selected from fibre links available to GÉANT, to accommodate a loss budget and link asymmetry compatible with the Twin Field QKD (TF-QKD) protocol. Nodes in the network are interconnected via fibre duplex cables, a typical resource in telecommunication infrastructure. Normally, they are used to implement full-duplex communication with each fibre carrying a unidirectional stream of data: one transmitting data (TX) and the other receiving data (RX).

In this experiment, each duplex includes one *classical fibre* (F_C) and one *quantum fibre*. While the classical fibre carries bidirectional data for the control layer, the quantum fibre transmits unidirectional encoded quantum (λ'_Q) and supports (λ'_S) signals from the transmitters to the central node. A comprehensive description of all the signals carried by the *classical fibres* is provided in section S3.

All fibres were characterised before the experiment via Optical Time-Domain Reflectometry (OTDR) and channel loss measurements. Table S1 outlines the losses and lengths of the deployed fibres. For each duplex, the fibre with the lowest losses was designated as the *quantum fibre*, to minimise the overall loss and enhance quantum communication performance.

Each fibre, F_Q^{AC} and F_C^{AC}, linking Frankfurt to Kirchfeld (Alice's and Charlie's locations, respectively), comprises two fibre spans joined in a colocation centre in Lampertheim, positioned approximately midway.

	Label	Attenuation (dB)	Purpose	Length (km)
Frankfurt ↔ Kirchfeld	F_Q^{AC}	32.4	Quantum Protocol	156.7
	F_C^{AC}	33.5	Classical Communications	
Kirchfeld ↔ Kehl	F_Q^{BC}	21.7	Quantum Protocol	97.2
	F_C^{BC}	22.2	Classical Communications	

TABLE S1. Properties of the deployed fibres used in the experiment.

S3. SIGNALS TRANSMITTED OVER THE CLASSICAL FIBRES

The *classical communication layer* includes two optical fibres connecting Charlie to Alice (F_C^{AC}) and to Bob (F_C^{BC}). These fibres carry several optical signals at different wavelengths travelling in two directions, upstream and down-

stream. The routing of the optical signals in the system is optimised for long-distance communication and is represented in fig. 2 of the main text, and in more detail in fig. S6. Upon reaching the designated receiving node, all transmitted signals are amplified using Erbium Doped Fibre Amplifiers (EDFAs), and signals at different optical frequencies are separated using Dense Wavelength Division Multiplexings (DWDMs). The optical wavelength and purpose of all the signals transmitted over the *classical fibres* are summarised in table S2.

ITU CH	**Wavelength (nm)**	**Purpose**	**Signal direction**	**Description**
CH32	1551.72	Protocol reference	C → A,B	CW optical reference for the quantum layer - for λ_Q
CH34	1550.12	Protocol signal	C → A,B	CW optical reference for the quantum layer - for λ_S
CH55	1533.47	FPGAs direct link (dwnlink)	C → A,B	Optical signal from $FPGA_C$ to $FPGA_{A,B}$ for direct FPGA communication and time synchronisation
CH56	1532.68	Ethernet downlink	C → A,B	Optical signal from Charlie's Ethernet switch to Alice and Bob
CH57	1531.90	FPGAs direct link (uplink)	A,B → C	Optical signal from $FPGA_{A,B}$ to $FPGA_C$ for direct FPGA communication and time synchronisation
CH58	1531.12	Ethernet uplink	A,B → C	Optical signals from Alice and Bob's Ethernet switches to Charlie

TABLE S2. Description of the optical signals transmitted over the classical fibres.

S4. PHASE NOISE COMPARISON BETWEEN FIBRE SPOOLS AND DEPLOYED FIBRES

As mentioned in the main text, preliminary tests on the deployed system were conducted on a lab-based testing rig in Cambridge. By utilising delay fibre spools and optical attenuators, the testing rig accurately emulated the primary characteristics of the installed fibres, namely their lengths and losses. However, as outlined in the Results subsection, some channel properties differed. Notably, as discussed in the main text, the polarisation drifts introduced by the deployed fibres were significantly lower than those observed in the testing rig. This difference is attributed to the higher temperature stability of the deployed buried fibres in comparison to the fibre spools utilised in the lab.

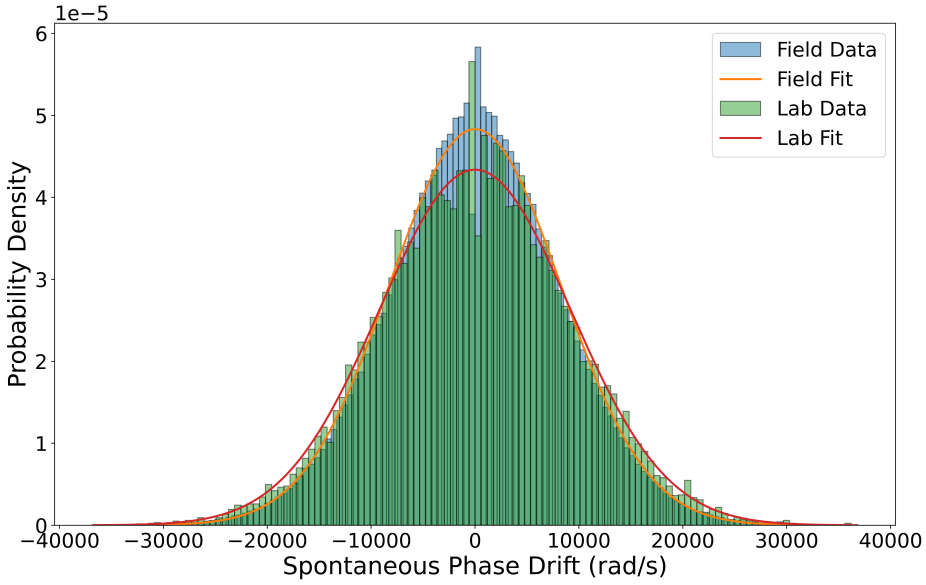


FIG. S1. Comparison of spontaneous phase drift between field and lab measurements. The normalised distribution of the spontaneous phase drift measured in the field (with the lab-based testing rig) is shown in light blue (green). Both distributions exhibit a Gaussian shape, consistent with previous observations. The normal curves are centred on 0 rad/s, with a standard deviation of 9197 rad/s for the lab data, and 8256 rad/s for the field data.

Another relevant property to compare between the two settings is the phase noise. To explore this, we acquired the free-drifting interference of the quantum signal both in the lab and the field, using our quantum communication

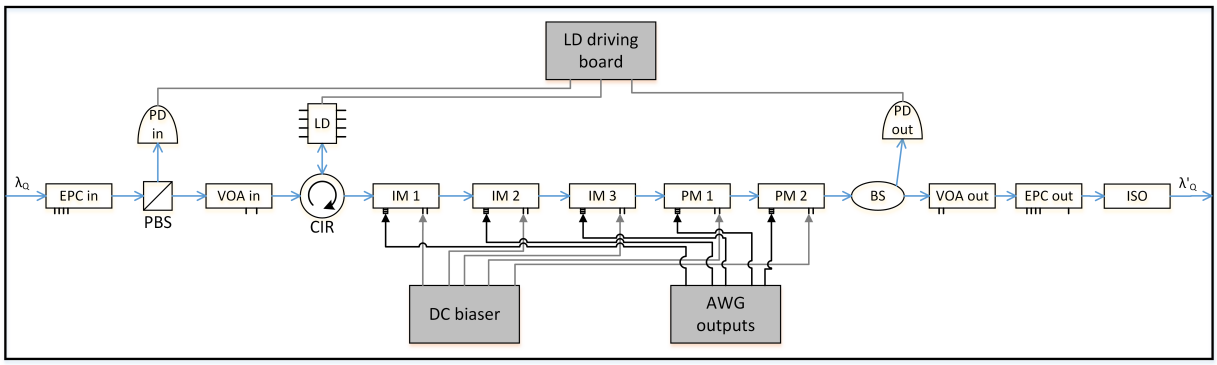


FIG. S2. Schematic of the Quantum Encoder unit endowed to Alice and Bob. The Quantum Encoder receives a CW λ_Q signal from the Quantum Signal MUX/DEMUX unit, and it encodes it with the TF-QKD protocol. The components inside the unit are: EPC: Electronic Polarisation Controller; PBS: Polarising Beam Splitter; PD: Photodiode; VOA: Variable Optical Attenuator; CIR: Circulator; LD: Laser Diode; IM: Intensity Modulator; PM: Phase Modulator; BS: Beam Splitter; ISO: Optical Isolator.

prototype and hence the same experimental setup. From this data, we extracted the distribution of the spontaneous phase drift rate and compared the two cases. Our analysis, shown in fig. S1, reveals that the phase drift rates measured in the field and the lab are very similar. Specifically, the standard deviation of the spontaneous phase drift measured in the lab is 9197 rad/s, while the same quantity in the installed fibres amounts to 8256 rad/s. This comparison serves as a valuable reference for the future, providing supporting evidence that the phase noise introduced by fibre spools in the lab effectively simulates that of installed fibres. It is worth noting that the phase drift measured in our experiment is comparable with the results obtained in previous experiments that used installed optical fibres.

S5. THE QUANTUM ENCODER

The Quantum Encoder is the unit in each transmitter node responsible for encoding the quantum communication protocol. It receives the Continuous Waveform (CW) λ_Q reference signal distributed by the *service layer* through the Quantum Signal MUX/DEMUX unit. This signal is processed by several optical modulators/components arranged in series to encode the SNS TF-QKD protocol. Figure S2 illustrates the internal components of the Quantum Encoder.

Light entering the Quantum Encoder first encounters an Electronic Polarisation Controller (EPC), a Polarisation Beam Splitter (PBS) and a photodiode (PD IN) configured to minimise the optical power recorded by PD IN. This mechanism compensates for polarisation rotations introduced by λ_Q distribution over classical fibres (F_C^{AC} and F_C^{BC}), aligning most of the received signals to the PBS's horizontal polarisation axis. This mechanism is described in more detail in section S9, and is one of the several automatic stabilisation routines implemented by the system.

The next set of components includes a Variable Optical Attenuator (VOA), followed by an optical Circulator (CIR), with port #2 connected to a Laser Diode (LD). These components implement an Optical Injection Locking (OIL) setup. The VOA attenuates the horizontally polarised component of incoming λ_Q exiting the PBS to ensure optimal injection locking of the laser diode cavity. This way, the laser diode's output is frequency-locked to the λ_Q signal, then returned to the CIR, and forwarded to ensuing modulators.

Five high-speed electro-optic modulators follow, applying necessary intensity and phase modulations for TF-QKD protocol implementation. Details on the encoded patterns are reported in section S7. To achieve a high 35 dB intensity contrast between the brightest and dimmest encoded pulses, we employ three Intensity Modulators (IMs) in series in the Quantum Encoder. The three IMs carve 120 ps-long optical pulses with four different intensities out of the CW signal received by the CIR. IM 1 is driven by a digital Radio Frequency (RF) signal, which carves the received CW signal by driving IM 1 between its maximum and minimum transmission settings. Without further intensity modulation, these carved pulses constitute the brightest pulses produced by the Encoders (s_A, u_A, and s_B). IM 2 is driven by a three-level RF signal and encodes the decoy states v_A, u_B, and v_B. IM 3 is driven by a digital RF signal, and increases the extinction ratio between the encoded and the vacuum pulses (w_A and w_B).

The intensity modulators are followed by two phase modulators. Phase Modulator (PM) 1 is driven by an analogue RF signal which encodes an arbitrary phase in the range $\left[-\frac{\pi}{2};+\frac{\pi}{2}\right)$. PM 2 is driven by a digital RF signal which encodes a phase offset of either $-\frac{\pi}{2}$ or $+\frac{\pi}{2}$. The combined effect of the two PMs allows the encoding of an arbitrary phase between $[-\pi;+\pi)$ onto the transmitted optical pulses.

Each of the five electro-optic modulators receives a dedicated DC biasing signal, optimizing intensity extinction

ratio for IMs and overall phase offset for PMs. An Arbitrary Waveform Generator (AWG) per transmitter node generates the five RF signals driving the modulators. These signals are carefully time-synchronised to the ps-level for optimal cascaded modulation.

A Beam Splitter (BS) then redirects most of the light to a second photodiode (PD out), monitoring average optical power. This photodiode is used to optimise the Direct Current (DC) bias of the preceding electro-optic modulators and monitor the average photon flux out of the transmitter.

The last components in the transmitter are VOA out, EPC out and an optical isolator. VOA out fine-tunes the intensity of the transmitted optical signal to the target average photon flux of the encoded pattern. EPC out optimises the polarisation of the transmitted signal to precompensate for polarisation rotations introduced by the quantum channels (F_Q^{AC} and F_Q^{BC}). The mechanism for this precompensation is described in more detail in section S9. Finally, to prevent optical signals such as back-reflections from entering the Quantum Transmitter through the output port, an optical isolator is installed as the final component of the Quantum Encoder.

At the output of the Quantum Encoder, the signal fully encoded with the quantum communication protocol is referred to as λ'_Q.

S6. IMPLEMENTATION SECURITY CONSIDERATIONS

As with any QKD system, the security of our implementation relies on ensuring a precise alignment between the practical realisation of the system and the theoretical assumptions underpinning its security proof [11]. Discrepancies between these two aspects can introduce security loopholes or side channels, potentially exploitable by an attacker to gain partial or full knowledge of the exchanged key without detection [12].

Over the past two decades, significant efforts have been made to identify and mitigate known vulnerabilities. The QKD community remains actively engaged in this endeavour, proposing countermeasures in the form of hardware and software updates or by refining security proofs to incorporate previously overlooked implementation nuances. These efforts aim to bridge the gap between theory and practice, closing in on the objective of delivering information-theoretic secure cryptographic capabilities.

In practice, both the transmitters and receivers of QKD systems are potential targets for attacks [13, 14], with receivers generally considered the most vulnerable element. A practical solution to eliminate all vulnerabilities associated with the receiver is to adopt a Measurement-Device-Independent (MDI) approach [15]. This method is inherently immune to detector-control attacks and removes all security requirements related to the receiver. In fact, in MDI-QKD, even if the receiver is fully controlled by an attacker, the security of the protocol remains uncompromised.

The TF-QKD protocol implemented in this work is a measurement-device-independent protocol. As such, it is inherently immune to all side channels associated with the receiver, which is located at the central node (Charlie). Consequently, to evaluate the security of our implementation, it is sufficient to assess the security of the transmitters, which is the focus of this section.

In the design of Alice's and Bob's encoders (fig. S2), all standard best practices for implementing QKD transmitters were applied [16]. From an implementation security perspective, the primary distinction between typical QKD transmitters and those used in our system lies in the need to phase-lock the local laser source ("LD" in fig. S2) to an external frequency reference. In our implementation, this is achieved through Optical Injection Locking (OIL). Special attention was therefore devoted to addressing potential vulnerabilities arising from this approach.

Attacks targeting or exploiting Optical Injection Locking (OIL) have been studied extensively over the years [17–22]. In an OIL attack, an adversary injects light into the transmitter with the aim of influencing the optical fields generated by the transmitter's laser source. These attacks aim to modify the properties of the prepared optical field—like phase, intensity, frequency, or polarisation—in a manner that deviates from the protocol's prescribed behaviour. Such manipulations allow the attacker to extract information from the transmitted pulses that would otherwise remain inaccessible.

Our security framework addresses two principal vulnerabilities of OIL: malicious injection through the quantum channel and via the service fibre.

To the best of our knowledge, all prior works on OIL attacks reported in the literature have focused exclusively on attacks conducted by injecting light through the output port of the transmitter (i.e., via the quantum channel). These studies have provided crucial insights into effective countermeasures to mitigate this threat, which informed our system design choices as detailed in later sections. The most effective strategy to defend against such attacks is to implement substantial optical isolation between the output port and the laser cavity, a measure that has been incorporated into our design.

Beyond these established countermeasures, we also proactively addressed the potential threat of OIL attacks carried out via the service channel, representing an alternative and less explored attack vector. While OIL has been used as

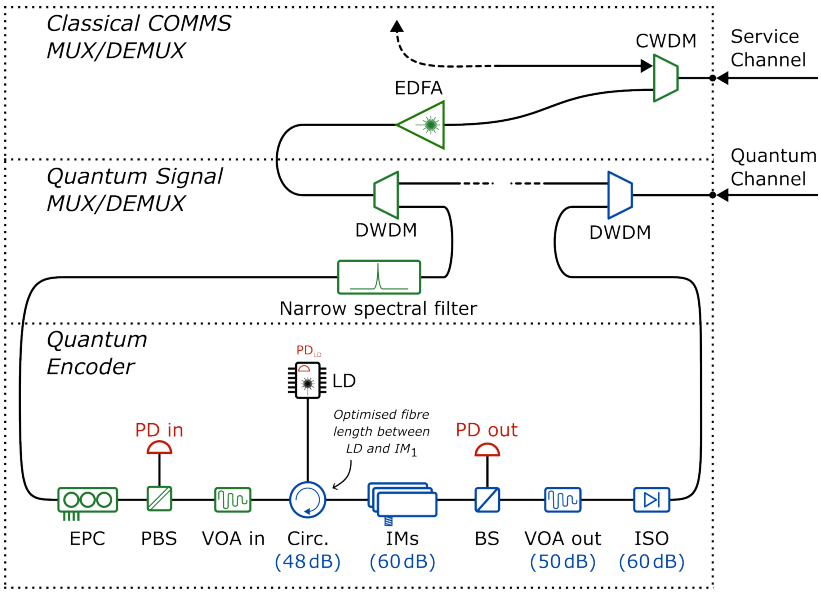


FIG. S3. Simplified schematic of the optical components traversed by an optical signal at λ_Q, injected into a transmitter node via the Service or Quantum Channel, before reaching the transmitter's local Laser Diode cavity. The scheme shows only the components relevant to the security considerations of OIL implementation. In green are the components traversed by light injected through the Service Channel. Red denotes the optical power probes integrated in the Encoder to monitor and maintain the injected and output power at the expected levels. In blue are represented components that attenuate optical signals injected through the Quantum Channel. Abbreviations: CWDM: Coarse Wavelength Division Multiplexer; EDFA: Erbium-Doped Fibre Amplifier; DWDM: Dense Wavelength Division Multiplexer; EPC: Electronic Polarisation Controller; PBS: Polarising Beam Splitter; PD: Photodiode; VOA: Variable Optical Attenuator; CIR: Circulator; LD: Laser Diode; IM: Intensity Modulator; BS: Beam Splitter; ISO: Optical Isolator.

a tool for phase-locking TF-QKD transmitters for several years now [23–25], no dedicated studies have thoroughly evaluated the vulnerabilities associated with this approach. Notably, even the only study explicitly targeting OIL in the context of TF-QKD, ref. [20], implemented the attack through the quantum channel.

To mitigate vulnerabilities arising from the service channel, we conducted an analysis of this attack vector. Our findings identified intensity manipulations as the primary threat to system security. To address this, we implemented several passive countermeasures to restrict direct access to the laser diode cavities of the transmitters to light injected from the service channel. Additionally, we developed a robust active monitoring system that continuously tracks and tune the power levels of both injected and transmitted signals, as described in detail in a later section.

Attacks via Quantum Channel

Attacks via the quantum channel, where light is injected into the encoder through its output port, are the most studied in literature [17–22]. They pose a significant threat, as the injected light can traverse the modulators responsible for qubit encoding and potentially reveal information about the prepared quantum states [18–21].

To counter this, prior studies indicate that sufficient attenuation between the LD and the transmitter's output port effectively shields the laser cavity from external injection locking [18–22]. According to [22], placing more than 140 dB of attenuation between the laser cavity and the output port of the transmitter ensures that the optical power an attacker would need to successfully inject the local LD exceeds the Laser-Induced-Damage Threshold (LIDT) of optical fibres. The LIDT, defined as the maximum power that an optical fibre can withstand without sustaining damage, is quoted in [26] as 55 kW for standard single-mode silica fibres. Beyond this threshold, the fibre softens and begins to melt.

The total attenuation in our system is greater than 200 dB, exceeding the threshold recommended in [26], thereby protecting the system from OIL attacks via the quantum channel. Below, we list the components in our transmitter that contribute to the overall isolation, starting from the most external-facing elements and moving toward the LD cavity. These components are also illustrated in fig. S3, which summarises all the countermeasures in place in our system to mitigate OIL attacks:

- A Dense Wavelength Division Multiplexing (DWDM) spectral filter located within the "Q. Signal MUX/DEMUX" module (figs. S3 and S6) blocks most of the injected power at wavelengths outside ITU CH34.
- A 60 dB optical isolator ("ISO" in figs. S2 and S3) positioned at the Encoder's output further blocks the majority of injected power.
- A combination of fixed and variable optical attenuators ("VOA out" in figs. S2 and S3) reduces any remaining injected light by at least 50 dB.
- In accordance with [20], we ensured that the time interval between the laser and the first intensity modulator (IM 1) is not a multiple of the pulse repetition rate. This design ensures that any injected light reaching the LD during quantum state preparation passes through the three IMs while they are in their blocking mode. This configuration provides an additional attenuation > 60 dB.
- Finally, a circulator further enhances isolation by > 46 dB, redirecting residual signals away from the laser cavity.

Together, these measures provide an overall attenuation greater than 200 dB, which largely exceeds the recommended 140 dB threshold recommended in literature [22, 26]. This ensures robust protection against OIL attacks via the quantum channel.

Attacks via the Service Fibre

In a QKD system employing OIL to lock the transmitter to an external frequency reference, an adversary could attempt to manipulate the light injected into the local laser diode via the service fibre. Given the lack of existing literature on such attacks, implementing effective countermeasures required a tailored analysis. The potential degrees of freedom an attacker might exploit include the polarisation, phase, or intensity of the prepared Weak Coherent Pulses (WCPs). Below we analyse these three degrees of freedom individually. Figure S3 includes also the representation of the countermeasures designed to defend against attacks conducted through the service channel.

Polarisation: To protect the transmitters from polarisation tampering, a Polarisation Beam Splitter (PBS) is placed as the first element in the encoder (labelled "PBS" in figs. S2 and S3). This ensures that only a polarisation mode can reach the LD in the transmitters, effectively mitigating potential manipulation

Phase: An attack that induces phase alterations in the prepared quantum state is equivalent to phase noise introduced by the transmission channel. Such noise is already accounted for in the security proof [27] and, therefore, does not compromise the security of the protocol.

In our TF-QKD protocol implementation, both Alice and Bob independently and privately apply phase randomisation to their Weak Coherent Pulses (WCPs) before encoding the phase and transmitting them. This randomisation process effectively erases any initial phase offset of the reference signal.

Indeed an OIL-based attack targeting the phase manipulation of LD encoders was proposed in [17]. That attack aimed to compromise the security of a QKD protocol by eliminating the phase randomisation of the generated WCPs, undermining the decoy state method. However, this attack is not applicable in our case. In our implementation, the phase randomisation of the WCPs is not reliant on the spontaneous phase fluctuations of the LD cavity; instead, it is actively achieved through the modulation of PM 1 (in fig. S2). As such, this type of attack is ineffective against our system.

Moreover, in an extreme scenario where an adversary introduces rapid phase fluctuations that exceed the compensation capabilities of the phase stabilisation mechanism, these fluctuations would manifest as an increased Quantum Bit Error Rate (QBER). This would render the attack detectable. Consequently, it is expected that such an attack would, at most, result in a Denial-of-Service (DoS) scenario, a common outcome in any QKD system when an adversary attempts to interfere with the transmitted qubits.

Intensity: Variations in intensity pose a significant risk, as fluctuations in the injected power could influence the output of the local LD, potentially compromising the implementation of the decoy-state method. To address this, we have first restricted direct access to the transmitters' LDs through spectral filtering and EDFA amplification, as shown in fig. S3. The spectral filtering ensures that only optical signals narrowly centred around ITU CH34 can reach the transmitter's LD, while the EDFA limits the maximum power that can reach the transmitter's LD.

These passive defenses are complemented by three active power-monitoring systems that continuously track both the injected and output power. A VOA positioned at the Encoder's input further enables fine-tuning of the injected power:

- A photodiode ("PD in" in figs. S2 and S3) continuously monitors changes in the injected power by detecting the orthogonal polarisation component. Due to the finite extinction ratio of the PBS in the encoder, this photodiode always receives a small fraction of the injected power, allowing it to effectively track any fluctuations in the injected signal.
- The local Laser Diode (LD) in the encoder (figs. S2 and S3) is equipped with a back-facet monitor detector. While this detector primarily ensures optimal phase locking of the LD, it also tracks power within the laser cavity. This provides an additional mechanism to detect anomalous fluctuations in the injected power, offering indicators of a potential OIL attack.
- A photodiode located at the Encoder's output ("PD out" in figs. S2 and S3) monitors the photon flux of the prepared signal. This ensures that the output power aligns with the expected levels, providing a further layer of verification to detect discrepancies that could indicate tampering or power instability.

These monitoring systems are integrated into a custom-built laser diode driving board ("LD driving board" in fig. S2), which actively tracks for tampering or fluctuations in injected power. Power readouts from these probes can be used to adjust the attenuation value set on the VOA at the entrance of the Encoder ("VOA in" in figs. S2 and S3), enabling precise control of the power injected into the transmitter's local LD. This comprehensive approach ensures that the intensity at the Encoder's output matches the desired photon flux, thereby mitigating potential attacks.

Final Considerations

While we have conducted a thorough analysis and implemented numerous countermeasures to ensure the security of adopting the OIL technique in our QKD implementation, we recognise that implementation security remains a highly active area of research. This is particularly true for techniques that have been adopted relatively recently in QKD systems. OIL was first introduced to QKD by [23] in 2019, and later adopted in [24, 25]. Despite several years having passed since its introduction, no dedicated implementation security studies have been conducted to assess its security implications, particularly with respect to attacks carried out through the service fibre.

We hope this will change in the future, with the research community thoroughly assessing potential vulnerabilities of this technique and proposing additional countermeasures to address them.

It is also important to note that, should the OIL technique reveal vulnerabilities in the future or prove to require countermeasures that are overly complex to implement, alternative frequency-locking techniques could be readily employed. One such method is frequency regeneration via an Optical Phase Locked Loop (OPLL), particularly its implementation achieving frequency locking by modulating the laser diode current. This approach, demonstrated in prior works [23, 28], offers comparable frequency-locking performance to OIL but comes with increased complexity due to the additional electronic components required. Similar to OIL, no studies to date have identified security vulnerabilities associated with adopting this OPLL method for frequency locking.

S7. ENCODING PARAMETERS

In this experiment, we implemented the 4-intensity SNS-TF-QKD quantum communication protocol [27, 29–31] across a highly asymmetric link. For the implementation, we used a 418800-pulse-long repeated encoded pattern. Interleaved with these are an equal number of phase unmodulated pulses, serving as reference signals to estimate the phase offset between Alice's and Bob's phase reference frames. The time-averaged outcome of their interference provides the error signal for the *fine* phase stabilisation feedback.

Pattern encoding is executed via two AWGs, one per transmitter node, synchronised remotely via an optical 125 MHz reference clock distributed from the central node to transmitter nodes through the classical fibres. Detailed information on the RF signals driving the high-speed electro-optic modulators is provided in section S5.

Pattern properties were selected to maximise Secret Key Rate (SKR) generation under our experimental conditions, considering channel losses and link asymmetry. Given the asymmetric configuration for this experiment, the pulses' intensities and sending probabilities are set to satisfy the security condition for asymmetric protocol execution outlined in the Methods. The chosen pattern properties are summarised in table S3.

Patterns at the transmitters were generated based on probabilities from table S3 through fair sampling. Initially, a 418800-elements-long list of pulse pairs matching these probabilities was created. Subsequently, the list was randomly shuffled and interleaved with the unmodulated phase referencing pulses. This approach avoids the effect of statistical fluctuations on pulse pairs distribution inherent in a repeated pattern generated through Monte Carlo approaches.

	Alice	Bob
s (ph/pulse)	0.52	0.24
u (ph/pulse)	0.52	0.13
v (ph/pulse)	0.08	0.012
w (ph/pulse)	0.0002	0.0002
P_Z (code basis)	80.0%	80.0%
$P_s(\epsilon)$	42.0%	15.0%
P_X (test basis)	20.0%	20.0%
P_u	5.0%	5.0%
P_v	80.0%	80.0%
P_w	15.0%	15.0%

TABLE S3. Parameters used in implementing the SNS TF-QKD protocol [27, 29]. s denotes the photon flux for signal pulses, while u, v, and w represent the photon fluxes for decoy states. P_Z is the users' probability of choosing the code basis, and P_s (ϵ) is the probability of sending a signal when the Z basis is chosen. P_X is the users' probability of encoding a pulse in the test basis, and P_u, P_v, and P_w are the probabilities of sending the decoy pulses u, v, w.

S8. THE QUANTUM RECEIVER AND DETECTORS UNITS

The Quantum Receiver combines signals received from Alice and Bob and extracts their orthogonal polarisation components, which are induced by polarisation rotations over the quantum channel. It then directs all the signals to four Avalanche Photo Diodes (APDs) for detection. Inputs are from the F_Q^{AC} and F_Q^{BC} fibres linking Charlie to Alice and Bob, respectively. Inside the receiver, signals first pass through two PBSs to align the polarisation of the incoming signals to the polarisation axes of the ensuing optical components. Vertical polarisation components are passed to a 4x1 optical switch, which dynamically selects and directs one to APD_4 for polarisation optimisation. The horizontal polarisation components of Bob's signals pass through a PM, which serves as an actuator for the *coarse* feedback mechanism. The horizontal polarisation components of Alice's and Bob's signals then interfere on a polarisation-maintaining BS. The interference outcomes are frequency demultiplexed by two DWDMs, with the outputs at λ_Q directed to APD_1 and APD_2. Outputs at λ_S are dynamically selected by a 2x1 optical switch and are directed to APD_3. Figure S4 for a schematic of the quantum receiver, and table S4 for component characterisation.

Component	Transmission
PBSs	93.0 %
PM BSs	93.7 %
DWDMs	87.3 %
Phase Modulator	64.6 %

TABLE S4. Transmissivity of the optical components inside the quantum receiver. The overall transmission of the receiver (excluding the PM) is measured to 76.0 %.

The APDs 1 to 4 are −30 °C-cooled 1 GHz-gated self-differenced APDs [32]. Critical to the successful execution of the quantum communication protocol is the selection of deadtime on time tagger used to analyse the APDs output signals. For APDs 1 and 2, which are used to monitor interference outcomes of protocol-encoded pulses, a long deadtime of 5 µs was set to effectively suppress afterpulse effect on secret key rate generation [33–35]. Conversely, for APDs 3 and 4, responsible for monitoring the signals supporting the protocol execution, a much shorter deadtime of 64 ns was chosen to maximise the maximum countrate on these detectors. The characterisation of all APDs is reported in table S5.

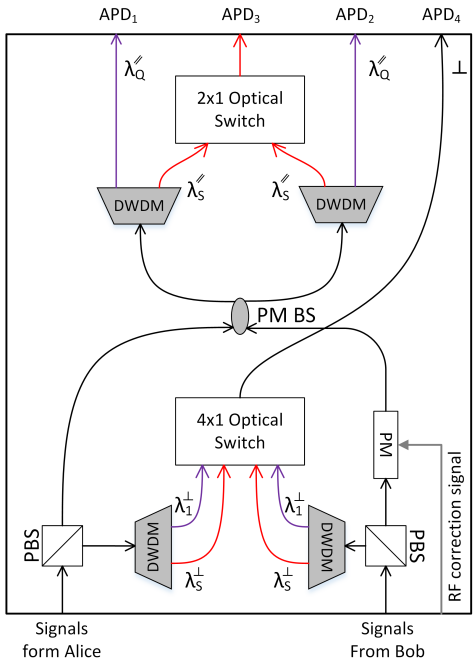


FIG. S4. Schematic of the Quantum Receiver unit. PM BS: Polarisation Maintaining Beam Splitter, PM: Phase Modulator, PBS: Polarising Beam Splitter, DWDM: Dense Wavelength Demultiplexer, APD: Avalanche Photodiode.

	APD_1 & APD_2	APD_3 & APD_4
Efficiency	14.5 %	15 %
Dark counts	450 Hz	500 Hz
Set dead-time	5 µs	64 ns

TABLE S5. Summary of the properties of the APDs used in the experiment.

S9. STABILISATION MECHANISMS

To support the continuous operation of the system over 7.5 h, several automatic optimisation mechanisms were implemented and are described below.

Time synchronisation of high-speed electronics: Time synchronisation of high-speed electronics (AWGs at the transmitters and time tagger at the receiver node) is achieved through a custom-developed system. Pairwise interconnected Field-Programmable Gate Arrays (FPGAs) at the nodes support the precise time synchronisation. Charlie's FPGA generates a 125 MHz local reference clock, which is distributed optically to Alice and Bob via SFPs and the classical fibres F_C^{AC} and F_C^{BC}. Fast PLLs synchronise the FPGAs, ensuring coordinated operation of all equipment. RF reference clocks and trigger signals from each FPGA synchronise the AWGs and time tagger.

Optimisation of the polarisation for the Quantum Encoder: The λ_Q reference signals distributed from Charlie to Alice and Bob undergo polarisation rotations introduced by the classical fibres F_C^{AC} and F_C^{BC}. To counter these rotations, each Quantum Encoder features a polarisation optimisation subsystem. This subsystem, comprised of the EPC in, PBS, and PD in components at the entrance of the Quantum Encoder (represented in fig. S2), continuously adjusts the settings of EPC using an automated algorithm. The algorithm minimises the power readout at PD in, ensuring alignment of the incoming light with the horizontal polarisation axis of the PBS and compensating for the polarisation rotations introduced by the classical fibres.

OIL optimisation: The OIL parameters for both transmitters are initially optimised in the lab to ensure optimal coherence transfer between the primary (L_S) and secondary (LD) laser sources. This optimisation includes adjusting the laser diode temperature, DC bias, and injected optical power. These parameters are replicated in the field. The injected power is fine-tuned to the optimal level by adjusting the VOA in (see fig. S2), guided

by the power reading from a photodiode within the LD cavity. Once the injected power reaches the optimal level, the DC bias and temperature of the LD are adjusted to the desired settings using a LD driving board (as depicted in fig. S2). Thanks to the stability of the reference λ_Q signal transmitted by the service layer and the efficacy of the polarisation optimisation routine for the incoming λ_Q signal, the OIL optimisation process only needs to be performed once at the beginning of the experiment.

DC bias optimisation for the high-speed electro-optic modulators: Initial DC bias optimisation of the electro-optic modulators was ensured before the experiment by varying the DC bias applied to each modulator while concurrently monitoring the transmitted power using PD out (see fig. S2). Each IM was individually optimized while all others were set to maximum transmission. During the experiment, an additional mechanism mitigate the spontaneous DC drift of the modulators. This mechanism relies on monitoring the extinction ratio between the various types of encoded pulses transmitted by one user when the other user is sending vacuum states. From these measurements made by Charlie, each user tune the DC biases of their IMs to maintain the desired intensity relation between the pulses.

Optimisation of the photon flux out of the transmitter nodes: Optimisation of the average intensity of the encoded patterns just out of the transmitter nodes to align them with the photon fluxes specified in table S3 is done through the power readout of PD out, and the attenuation of the outgoing signal through VOA out. During the testing phase of the system, the intensity ratio between the signal output recorded by PD out (see fig. S2) and the signal injected into the quantum fibre is carefully characterised. This ratio remains fixed and can only be adjusted by additional attenuation provided by VOA out. As a result, the power measured at PD out serves as an indicator of the average optical intensity of the λ'_Q signal produced by the transmitter node. The PD out readout, alongside the knowledge of the encoded pattern properties, enables Alice and Bob to set their photon fluxes in alignment with those specified in table S3, ensuring adherence to the security requirements of the protocol implementation. The power readout of PD out is continuously monitored throughout the protocol execution, and the attenuation of VOA out is fine-tuned to maintain consistency with the photon fluxes outlined in table S3.

Optimisation of λ'_Q and λ'_S polarisation: Similarly to the polarisation rotations introduced by the classical fibres, which are addressed by the polarisation stabilisation routine implemented by the Quantum Encoder, the polarisation rotation introduced by the quantum fibres F_Q^{AC} and F_Q^{BC} also needs to be compensated for. In this case, compensation is applied at the transmitting end of the fibre rather than the receiving end. This configuration is chosen to minimise losses at the quantum receiver and enhance protocol performance. Precompensation is applied by the EPC out in fig. S2 for λ'_Q and by an EPC in the Quantum Signal MUX/DEMUX unit (seefig. S6) for λ'_S.

The polarisation optimisation mechanism adjusts the polarisation of all signals reaching the Quantum Receiver by minimising in turn the number of counts recorded by APD 4 for each signal directed to it by the 4x1 optical switch (see fig. S2).

Off-band phase stabilisation: Due to the critical role played by the off-band phase stabilisation in the protocol execution, a detailed description of this mechanism is outlined in the Methods subsection of the manuscript.

S10. SYSTEM'S PERFORMANCE IMPROVEMENTS BEYOND STATE-OF-THE-ART CRYOGENIC POINT-TO-POINT QKD

As mentioned in the main text, the system's performance could be further improved by cooling the APDs below $-30\,^{\circ}\mathrm{C}$, which has been demonstrated feasible with thermoelectric coolers [36]. By operating at $-60\,^{\circ}\mathrm{C}$, we estimate a reduction in the dark count rate by 10 dB while retaining the same detection efficiency. This improvement would significantly extend the system's range; with the TF-QKD system's SKR scaling with the square root of channel loss, the operational range could increase by 20 dB. This would raise the link loss budget from 56.0 dB to 76.0 dB, surpassing the state-of-the-art link loss of 71.9 dB reported for cryogenically cooled point-to-point QKD systems [37].

S11. NETWORK EXTENSION AND ALTERNATIVE PHASE STABILISATION METHODS

The star-shaped topology of our system naturally supports the expansion of network users. Each new user would have their own transmitter node, which includes all necessary components across the *service*, *management*, and

quantum layers, and would connect to the central node (Charlie) via a fibre duplex. One fibre in the duplex is used as service fibre, while the other is used as quantum fibre.

Expanding the number of service fibres connected to the central node poses no issue, as all communication over these fibres involves classical signals (i.e., high optical intensity). Signals transmitted over this fibre can be handled as follows:

- Amplified and distributed to reach more users for λ_S and λ_Q.
- Enhanced by adding equipment to incorporate new users into the Local Area Network (Ethernet downlink and uplink).
- Increased by expanding the number of direct links supported by the central FPGA (FPGA direct uplink and downlink).

Expanding the number of quantum fibres connected to the central node requires more consideration due to the quantum nature of the signals transmitted. In a network with three or more transmitters, a mechanism is needed to enable interference between signals from two different users at the time at the central node. The simplest solution is to equip the central node with an optical switch. This switch would allow interference between two users at a time, ensuring the successful execution of the protocol. In this configuration, key generation would not occur continuously among all users but would operate in a round-robin manner, coordinated by a Key Management System (KMS) [38].

A critical factor for a successful key generation when adding new users is ensuring that the phase stabilisation system functions effectively in the presence of distance asymmetries between users connected to the central node. Asymmetries in the optical paths of the signals interfering at the central node can introduce phase noise from the lasers, disrupting phase stabilisation. To prevent this, the optical path difference between interfering signals at Charlie must remain within the coherence length of the reference lasers. This can be achieved by using narrow-linewidth lasers at the central node, with a coherence length that exceeds the network's asymmetry. The star-shaped architecture of our system is well-suited to addressing this challenge efficiently. In our system, reference frequencies are distributed from the central node to all connected transmitters, meaning that narrow-linewidth lasers are required only at the central node, where they can be shared. This makes the system both scalable and practical, optimising performance and resource utilisation.

Implementation variations of the phase stabilsation scheme presented in this work can be envisioned. One promising direction is exploring other non-cryogenic single-photon detection techniques for the detectors in Charlie, such as silicon-based up-conversion single-photon detectors [39, 40]. These detectors leverage nonlinear optics to convert infrared photons into visible-range photons, enabling the use of highly sensitive silicon-based detection systems. This approach offers comparable dark count rates and detection efficiencies to the InGaAs detectors typically used for single-photon detection at 1550 nm.

Another avenue worth exploring is implementing the fast phase feedback for λ_S using highly sensitive photodiodes instead of Single Photon Detectors (SPDs) [41]. Since no quantum information is encoded on λ_S, single-photon detection is not strictly necessary for this wavelength. In the current implementations, SPDs are used to maintain a low-intensity signal on λ_S, thereby minimising the contamination of λ_Q due to nonlinear scattering effects. By contrast, using photodiodes for the fast stabilisation would likely require a higher-intensity signal on λ_S, which could increase noise on λ_Q. This approach may be well-suited for mid-range operations, though it could be less effective for long-distance communication, where maintaining a high signal-to-noise ratio is crucial.

Finally, a passive phase referencing mechanism could be envisioned, utilizing non-cryogenically cooled single-photon detectors and off-band stabilisation. For this approach to be effective, a precise relationship between the wavelengths used in the setup must be established, such as by generating the frequencies through a frequency comb. Once this is achieved, the phase noise information recorded by detectors monitoring the interference at λ'_S and λ'_Q could be correlated. Combined with a form of time-multiplexed phase referencing referencing scheme over λ'_Q [42], this should allow for accurate prediction of the instantaneous phase offset of the encoded pulses interfering at λ'_Q.

S12. COMPARISON WITH FIELD TRIAL EXPERIMENTS USING CRYOGENIC DETECTORS

To provide a comprehensive overview of the state-of-the-art QKD experiments over deployed fibres, fig. S5 presents the same data shown in fig. 5 of the main text, with additional experiments that employ cryogenically-cooled single-photon detectors (Superconductive Nanowire Single Photon Detectorss (SNSPDs)). For a fair comparison, all experimental data, simulations, and Secret Key Capacity bounds (SKCs) in this figure are plotted as a function of

channel loss. The equivalent channel length is indicated at the top of the graph, assuming a loss of 0.22 dB/km, which corresponds to the value observed in this experiment and represents a realistic loss coefficient for deployed fibres.

It is important to note that while the additional experimental points in fig. S5 were obtained from quantum signals transmitted over deployed optical fibres, none of these experiments was conducted with a system fully integrated into a telecommunication infrastructure. To the best of our knowledge, all results were obtained with part or all of the equipment installed in dedicated laboratories, not alongside operational telecommunications equipment. This highlights the challenges of deploying cryogenically cooled equipment in telecom environments.

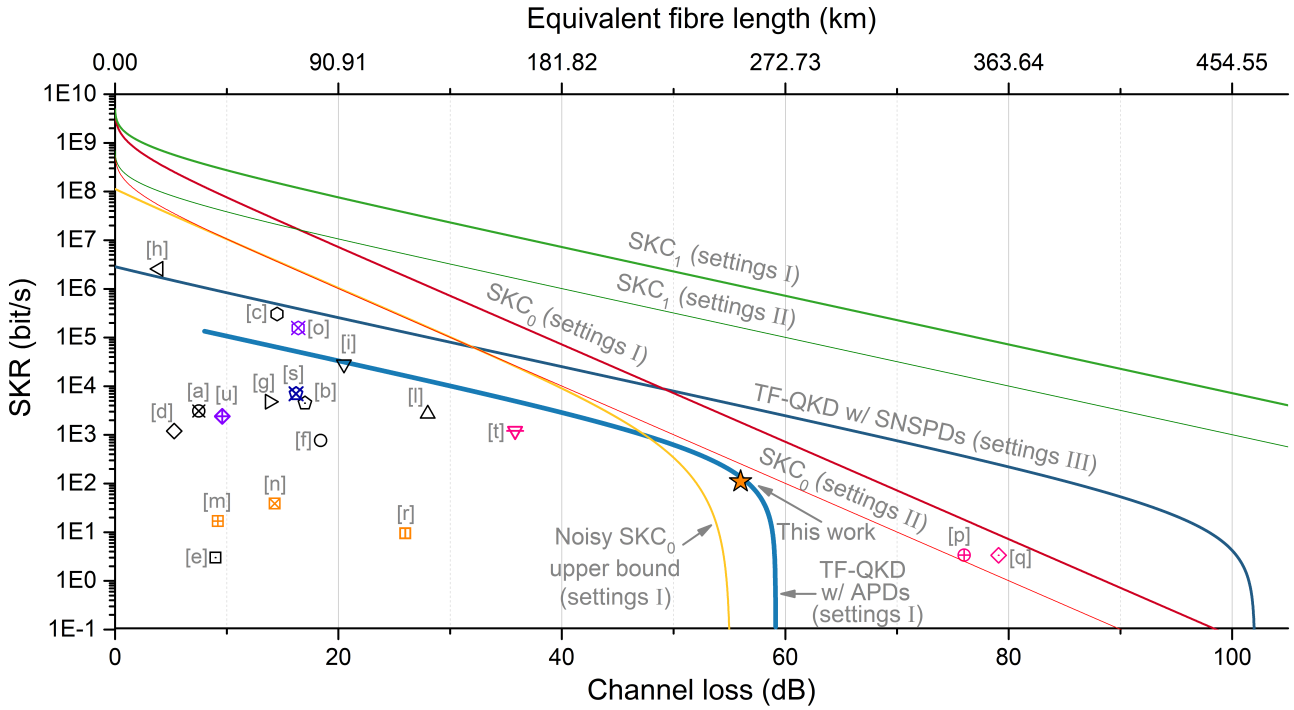


FIG. S5. Experimental results already reported in fig. 5 of the main text with an added comparison with QKD experiments using cryogenically-cooled equipment and transmitting signals over deployed fibres. The experimental markers are linked to their corresponding references as follows: [a] → [1], [b] → [2], [c] → [3], [d] → [4], [e] → [5], [f] → [6], [g] → [7], [h] → [8], [i] → [9], [l] → [10]. [m] → [43], [n] → [44], [o] → [45], [p] → [46], [q] → [25], [r] → [47], [s] → [48], [t] → [49], [u] → [50]. The colours of the added markers indicate the type of QKD protocol implemented in each experiment. The parameters for Settings I and II, used in the calculation of SKC_0 and SKC_1, are provided in table S6. The parameters for the TF-QKD protocol employing using cryogenically-cooled detectors (Settings III) are detailed in table S7.

The additional experiments plotted in fig. S5 are listed chronologically and linked to their respective references in the caption. The markers colour indicate the different QKD protocols implemented:

- Purple → point-to-point protocol, (e.g., BB84): [45, 50].
- Blue → entanglement-based protocol, (e.g., BBM92): [48].
- Orange → MDI-QKD: [43, 44, 47].
- Violet → Post-Measurement-Pairing QKD ([51]) and TF-QKD: [25, 46, 49].

Notably, this comparison shows that our experiment achieves approximately 75 % of the maximum distance (normalised by the fibre attenuation coefficient, the "Equivalent fibre length" axes in fig. S5) ever reached by a TF-QKD experiment over deployed fibres [25]. This result is particularly compelling given the substantial differences between the two technologies outlined in table I of the main text, including APDs' order-of-magnitude disadvantage in quantum efficiency, their two orders of magnitude higher dark counts, and their two orders of magnitude advantage in cost.

In fig. S5, alongside the markers for the experimental data, the SKC_0 and SKC_1 bounds are plotted for two experimental settings (see table S6). The lines for Setting I represent the bounds for an implementation using an encoding repetition rate of 0.5 GHz, which is the rate one used for this experiment. The lines for Setting II represent

the bounds for an implementation with an encoding repetition rate of 70 MHz. These latter bounds are included to provide a fair representation of the results from [25, 46], both of which exceeded their respective SKC_0 bound when scaled by the experiments' encoding rate (as shown in fig. S5).

The thin blue line in fig. S5 represent the expected SKR for the implementation of the TF-QKD protocol over a symmetric link using SNSPDs detectors. The parameters for the SKR simulation are detailed in table S7. This curve illustrates the SKR performance that our system could achieve if SNSPDs detectors were employed at the receiving node.

	Encoding repetition rate	**Other parameters**
Settings I	0.5 GHz	As per Main and Supplementary Materials
Settings II	70 MHz	—

TABLE S6. Details of Settings I and II in fig. S5. Settings I are the ones used in this experiment. Settings II are extracted from [25].

Encoding repetition rate	0.5 GHz
Detectors' efficiency	75 %
Detectors' dark counts	14 Hz
s	0.45 (ph/pulse)
u	0.3 (ph/pulse)
v	0.06 (ph/pulse)
w	0.0002 (ph/pulse)
P_Z (code basis)	0.8 %
P_s (ϵ)	0.28 %
P_X (test basis)	0.2 %
P_u	0.1 %
P_v	0.8 %
P_w	0.1 %
Other parameters	As per Main and Supplementary Materials

TABLE S7. Parameters for Settings III in fig. S5. This configuration corresponds to the SNS TF-QKD protocol [27, 29] where photon fluxes and pattern properties were optimised for a symmetric layout (i.e., $L_A = L_B$) and executed by a system employing SNSPDs cryogenic detectors. The detector parameters were derived from the authors' previous experimental work conducted in a laboratory setting [28].

S13. DETAILED SCHEMATIC OF THE QUANTUM COMMUNICATION SYSTEM

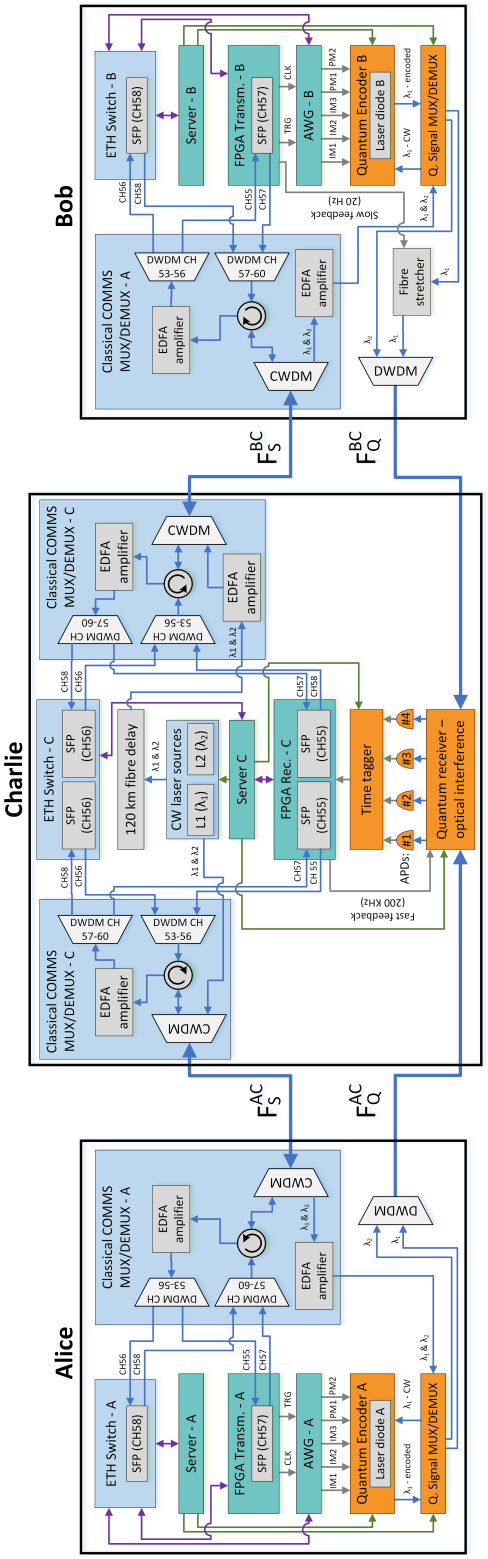


FIG. S6. Detailed representation of the developed coherent quantum communication system.

S14. DETAILED EXPERIMENTAL RESULTS FOR SNS TF-QKD PROTOCOL

Table S8 summarises the detailed experimental results obtained from the implementation of the SNS TF-QKD coherent quantum communication protocol between Frankfurt and Kehl. *N total sent* is the total number of encoded quantum pulses sent throughout the protocol execution. The number of pulses sent in each Alice-Bob pulses pair configuration can be calculated by multiplying the respective configuration probabilities (deducible from table S3) by *N total sent*. The number of pulses detected in these configurations are reported in table S8 with format "Detected-B1B2t1t2", where B1 and t1 (B2 and t2) are the basis and the type of pulse sent by Alice (Bob).

Fibre Length (km)	253.9
Alice - Charlie (km)	156.7
Bob - Charlie (km)	97.2
N total sent	1.36581×10^{13}
Phase mismatch acceptance	22.5°
Z error rate (before AOPP)	30.7%
Z error rate (after AOPP)	3.56%
Xuu error rate	4.61%
Xvv error rate	5.83%
Phase error rate	5.08%
Detected-ZZss	80342420
Detected-ZZsn	104894262
Detected-ZZns	86381667
Detected-ZZnn	4163565
Detected-ZXsu	4346359
Detected-ZXsv	28080211
Detected-ZXsw	4647897
Detected-ZXnu	3924498
Detected-ZXnv	3299312
Detected-ZXnw	180031
Detected-XZus	2405448
Detected-XZun	3107519
Detected-XZvs	31252233
Detected-XZvn	8882151
Detected-XZws	5576883
Detected-XZwn	261425
Detected-XXuu	129224
Detected-XXuv	841061
Detected-XXuw	136832
Detected-XXvu	1467103
Detected-XXvv	3873980
Detected-XXvw	392002
Detected-XXwu	249794
Detected-XXwv	418392
Detected-XXww	11671
SKC0 (bit/channel use)	$3.62 \cdot 10^{-6}$
SKC0 (bit/s)	1812.0
Number of secure bits generated (bits)	$3.01 \cdot 10^{6}$
SKR (bit/signal)	$2.20 \cdot 10^{-7}$
SKR (bit/s)	110.1

TABLE S8. Detailed results obtained from the SNS TF-QKD protocol implementation between Frankfurt and Kehl.

[1] Peev, M. *et al.* The secoqc quantum key distribution network in vienna. *New Journal of Physics* **11**, 075001 (2009).

[2] Chen, T.-Y. *et al.* Metropolitan all-pass and inter-city quantum communication network. *Optics Express* **18**, 27217–27225 (2010).

[3] Sasaki, M. *et al.* Field test of quantum key distribution in the tokyo qkd network. *Optics Express* **19**, 10387–10409 (2011).

[4] Stucki, D. *et al.* Long-term performance of the swissquantum quantum key distribution network in a field environment. *New Journal of Physics* **13**, 123001 (2011).

[5] Rubenok, A., Slater, J. A., Chan, P., Lucio-Martinez, I. & Tittel, W. Real-world two-photon interference and proof-of-principle quantum key distribution immune to detector attacks. *Physical Review Letters* **111**, 130501 (2013).

[6] Wang, S. *et al.* Field and long-term demonstration of a wide area quantum key distribution network. *Optics Express* **22**, 21739–21756 (2014).

[7] Mao, Y. *et al.* Integrating quantum key distribution with classical communications in backbone fiber network. *Optics Express* **26**, 6010–6020 (2018).

[8] Dynes, J. F. *et al.* Cambridge quantum network. *npj Quantum Information* **5**, 2728 (2019).

[9] Chen, Y.-A. *et al.* An integrated space-to-ground quantum communication network over 4,600 kilometres. *Nature* **589**, 214–219 (2021).

[10] Wonfor, A. *et al.* Quantum networks in the uk. In Glick, M., Srivastava, A. K. & Akasaka, Y. (eds.) *Metro and Data Center Optical Networks and Short-Reach Links IV*, 4 (SPIE, 06/03/2021 - 12/03/2021).

[11] Zapatero, V., Navarrete, Á. & Curty, M. Implementation security in quantum key distribution. *Advanced Quantum Technologies* 2300380 (2024).

[12] Lydersen, L. *et al.* Hacking commercial quantum cryptography systems by tailored bright illumination. *Nature Photonics* **4**, 686–689 (2010). Num Pages: 4.

[13] Pirandola, S. *et al.* Advances in quantum cryptography. *Advances in Optics and Photonics* **12**, 1012 (2020).

[14] Xu, F., Ma, X., Zhang, Q., Lo, H.-K. & Pan, J.-W. Secure quantum key distribution with realistic devices. *Reviews of Modern Physics* **92**, 131 (2020).

[15] Lo, H.-K., Curty, M. & Qi, B. Measurement-device-independent quantum key distribution. *Physical Review Letters* **108**, 130503 (2012).

[16] Dixon, A. R. *et al.* Quantum key distribution with hacking countermeasures and long term field trial. *Scientific Reports* **7**, 1978 (2017).

[17] Sun, S.-H. *et al.* Effect of source tampering in the security of quantum cryptography. *Physical Review A* **92**, 325 (2015).

[18] Huang, A. *et al.* Laser-seeding attack in quantum key distribution. *Phys. Rev. Applied* **12**, 073 (2019).

[19] Pang, X.-L. *et al.* Hacking quantum key distribution via injection locking. *Phys. Rev. Applied* **13**, 073 (2020).

[20] Zhang, X.-X. *et al.* Analysis of an injection-locking-loophole attack from an external source for quantum key distribution. *Phys. Rev. A* **106** (2022).

[21] Ponosova, A. *et al.* Protecting fiber-optic quantum key distribution sources against light-injection attacks. *PRX Quantum* **3** (2022).

[22] Lovic, V., Marangon, D. G., Smith, P. R., Woodward, R. I. & Shields, A. J. Quantified effects of the laser-seeding attack in quantum key distribution. *Physical Review Applied* **20** (2023).

[23] Minder, M. *et al.* Experimental quantum key distribution beyond the repeaterless secret key capacity. *Nature Photonics* **13**, 334–338 (2019).

[24] Fang, X.-T. *et al.* Implementation of quantum key distribution surpassing the linear rate-transmittance bound. *Nature Photonics* **14**, 422–425 (2020).

[25] Liu, H. *et al.* Field test of twin-field quantum key distribution through sending-or-not-sending over 428 km. *Physical Review Letters* **126** (2021).

[26] Lucamarini, M. *et al.* Practical security bounds against the trojan-horse attack in quantum key distribution. *Physical Review X* **5**, 325 (2015).

[27] Jiang, C., Hu, X.-L., Xu, H., Yu, Z.-W. & Wang, X.-B. Zigzag approach to higher key rate of sending-or-not-sending twin field quantum key distribution with finite-key effects. *New Journal of Physics* **22**, 053048 (2020).

[28] Pittaluga, M. *et al.* 600-km repeater-like quantum communications with dual-band stabilization. *Nature Photonics* **15**, 530–535 (2021).

[29] Wang, X.-B., Yu, Z.-W. & Hu, X.-L. Twin-field quantum key distribution with large misalignment error. *Physical Review A* **98** (2018).

[30] Hu, X.-L., Jiang, C., Yu, Z.-W. & Wang, X.-B. Sending-or-not-sending twin-field protocol for quantum key distribution with asymmetric source parameters. *Physical Review A* **100** (2019).

[31] Xu, H., Yu, Z.-W., Jiang, C., Hu, X.-L. & Wang, X.-B. Sending-or-not-sending twin-field quantum key distribution: Breaking the direct transmission key rate. *Physical Review A* **101** (2020).

[32] Yuan, Z. L., Dixon, A. R., Dynes, J. F., Sharpe, A. W. & Shields, A. J. Gigahertz quantum key distribution with ingaas avalanche photodiodes. *Applied Physics Letters* **92** (2008).

[33] Hadfield, R. H. Single-photon detectors for optical quantum information applications. *Nature Photonics* **3**, 696–705 (2009).

[34] Zhang, J., Itzler, M. A., Zbinden, H. & Pan, J.-W. Advances in ingaas/inp single-photon detector systems for quantum communication. *Light: Science & Applications* **4**, e286–e286 (2015).

[35] Humer, G. *et al.* A simple and robust method for estimating afterpulsing in single photon detectors. *Journal of Lightwave Technology* **33**, 3098–3107 (2015).
[36] Fröhlich, B. *et al.* Long-distance quantum key distribution secure against coherent attacks. *Optica* **4**, 163 (2017).
[37] Boaron, A. *et al.* Secure quantum key distribution over 421 km of optical fiber. *Physical Review Letters* **121**, 190502 (2018).
[38] James, P., Laschet, S., Ramacher, S. & Torresetti, L. Key management systems for large-scale quantum key distribution networks. In *Proceedings of the 18th International Conference on Availability, Reliability and Security*, ARES '23 (Association for Computing Machinery, New York, NY, USA, 2023).
[39] Shentu, G.-L. *et al.* Ultralow noise up-conversion detector and spectrometer for the telecom band. *Optics Express* **21**, 13986–13991 (2013). Publisher: Optica Publishing Group.
[40] Ma, F. *et al.* Upconversion single-photon detectors based on integrated periodically poled lithium niobate waveguides. *JOSA B* **35**, 2096–2101 (2018). Publisher: Optica Publishing Group.
[41] Clivati, C. *et al.* Coherent phase transfer for real-world twin-field quantum key distribution. *Nature Communications* **13**, 157 (2022).
[42] Liu, Y. *et al.* Experimental twin-field quantum key distribution over 1000 km fiber distance. *Physical Review Letters* **130**, 210801 (2023).
[43] Tang, Y.-L. *et al.* Field test of measurement-device-independent quantum key distribution. *IEEE Journal of Selected Topics in Quantum Electronics* **21**, 116–122 (2015).
[44] Tang, Y.-L. *et al.* Measurement-device-independent quantum key distribution over untrustful metropolitan network. *Physical Review X* **6**, 83 (2016).
[45] Bunandar, D. *et al.* Metropolitan quantum key distribution with silicon photonics. *Physical Review X* **8** (2018).
[46] Chen, J.-P. *et al.* Twin-field quantum key distribution over a 511 km optical fibre linking two distant metropolitan areas. *Nature Photonics* **299**, 1476 (2021).
[47] Berrevoets, R. C. *et al.* Deployed measurement-device independent quantum key distribution and bell-state measurements coexisting with standard internet data and networking equipment. *Communications Physics* **5**, 9288 (2022).
[48] Pelet, Y. *et al.* Operational entanglement-based quantum key distribution over 50 km of field-deployed optical fibers. *Physical Review Applied* **20** (2023).
[49] Zhu, H.-T. *et al.* Field test of mode-pairing quantum key distribution. *Optica* **11**, 883 (2024).
[50] Zahidy, M. *et al.* Quantum key distribution using deterministic single-photon sources over a field-installed fibre link. *npj Quantum Information* **10**, 1308 (2024).
[51] Avesani, M. Long-range quantum cryptography gets simpler. *Physics* **16**, 104 (2023).